

$$\text{The } w_0 \text{ character identity:}$$

$$\sum_{T \in \text{SYT}(\lambda)} (-1)^{\text{comaj}(T)} = \chi^\lambda(w_0)$$

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Abstract

For every partition $\lambda \vdash n$ I prove the closed-form identity

$$\sum_{T \in \text{SYT}(\lambda)} (-1)^{\text{comaj}(T)} = \chi^\lambda(w_0),$$

where $w_0 \in S_n$ is the longest element (reversal $i \mapsto n+1-i$, of cycle type $2^{\lfloor n/2 \rfloor} 1^{n \bmod 2}$) and $\text{comaj}(T) = \sum_{i \in \text{Des}(T)} (n-i)$. This is the “central character identity” left open in the companion note *The $q \rightarrow -1$ fiber of the branch-exponent bridge*; closing it makes the $q = -1$ fiber of Theorem B a complete, self-contained theorem. The proof is short and structural: evacuation reduces the comajor sum to the fake-degree polynomial $f^\lambda(q) = \sum_T q^{\text{maj}(T)}$; Stanley’s principal specialization writes $f^\lambda(q) = (q; q)_n s_\lambda(1, q, q^2, \dots)$; the Frobenius character formula expands this over power sums; and the zeros of $(q; q)_n$ at $q = -1$ act as a *sieve* that annihilates every cycle type except w_0 ’s, with surviving constant exactly z_{w_0} . All steps are verified computationally for every partition of $n \leq 10$.

1 Statement and conventions

Throughout, $\lambda \vdash n$ is a partition of $n \geq 1$ and $\text{SYT}(\lambda)$ is the set of standard Young tableaux of shape λ . For $T \in \text{SYT}(\lambda)$ the *descent set* is

$$\text{Des}(T) = \{i \in [n-1] : i+1 \text{ lies in a strictly lower row of } T \text{ than } i\},$$

and we set

$$\text{maj}(T) = \sum_{i \in \text{Des}(T)} i, \quad \text{comaj}(T) = \sum_{i \in \text{Des}(T)} (n-i).$$

Let χ^λ denote the irreducible S_n -character indexed by λ , and for a partition $\mu \vdash n$ write $z_\mu = \prod_{r \geq 1} r^{m_r(\mu)} m_r(\mu)!$, where $m_r(\mu)$ is the number of parts of μ equal to r . The longest element $w_0 \in S_n$ is a fixed-point-free involution for n even and has a single fixed point for n odd; in either case its cycle type is

$$\mu_0 := 2^{\lfloor n/2 \rfloor} 1^{n \bmod 2}.$$

Theorem 1. *For every partition $\lambda \vdash n$,*

$$\sum_{T \in \text{SYT}(\lambda)} (-1)^{\text{comaj}(T)} = \chi^\lambda(w_0) = \chi^\lambda(\mu_0).$$

We prove Theorem 1 through four lemmas. Define the *fake-degree polynomial*

$$f^\lambda(q) := \sum_{T \in \text{SYT}(\lambda)} q^{\text{maj}(T)} \in \mathbb{Z}[q].$$

2 Reduction to the fake degree at $q = -1$

Lemma 1 (Evacuation: comaj equidistributes with maj). *For every $\lambda \vdash n$,*

$$\sum_{T \in \text{SYT}(\lambda)} q^{\text{comaj}(T)} = \sum_{T \in \text{SYT}(\lambda)} q^{\text{maj}(T)} = f^\lambda(q).$$

In particular $\sum_T (-1)^{\text{comaj}(T)} = f^\lambda(-1)$.

Proof. Let $\text{evac}: \text{SYT}(\lambda) \rightarrow \text{SYT}(\lambda)$ be the Schützenberger involution (evacuation). A classical theorem of Schützenberger states that evacuation complements the descent set in the sense

$$\text{Des}(\text{evac}(T)) = \{n - i : i \in \text{Des}(T)\}$$

(see Stanley, *Enumerative Combinatorics* vol. 2, Appendix A1, or van Leeuwen’s treatment of the Robinson–Schensted–Schützenberger correspondence). Hence

$$\text{maj}(\text{evac}(T)) = \sum_{j \in \text{Des}(\text{evac}(T))} j = \sum_{i \in \text{Des}(T)} (n - i) = \text{comaj}(T).$$

Since evac is a bijection of $\text{SYT}(\lambda)$, reindexing the sum gives $\sum_T q^{\text{maj}(T)} = \sum_T q^{\text{maj}(\text{evac}(T))} = \sum_T q^{\text{comaj}(T)}$. $\square$

Thus Theorem 1 is equivalent to the single evaluation

$$f^\lambda(-1) = \chi^\lambda(\mu_0). \tag{1}$$

3 Principal specialization and the Frobenius expansion

We work in the ring Λ of symmetric functions, with s_λ the Schur functions, p_μ the power sums, and the *geometric alphabet* $X_q = (1, q, q^2, q^3, \dots)$, on which power sums evaluate as the convergent geometric series (for $|q| < 1$, equivalently as rational functions of q)

$$p_r(X_q) = \sum_{j \geq 0} q^{rj} = \frac{1}{1 - q^r}, \quad r \geq 1. \tag{2}$$

Write $(q; q)_n = \prod_{i=1}^n (1 - q^i)$.

Lemma 2 (Stanley’s principal specialization). *For every $\lambda \vdash n$,*

$$f^\lambda(q) = \sum_{T \in \text{SYT}(\lambda)} q^{\text{maj}(T)} = (q; q)_n s_\lambda(1, q, q^2, \dots) = q^{n(\lambda)} \frac{(q; q)_n}{\prod_{x \in \lambda} (1 - q^{h(x)})},$$

where $n(\lambda) = \sum_i (i - 1)\lambda_i$ and $h(x)$ is the hook length of the cell x .

Proof. This is the standard principal-specialization formula for Schur functions; see Stanley, *EC2*, Corollary 7.21.2 and Theorem 7.21.2 (the hook-content/principal specialization $s_\lambda(1, q, q^2, \dots) = q^{n(\lambda)} / \prod_x (1 - q^{h(x)})$). We recall the descent half, which is what we use. By Gessel's fundamental-quasisymmetric expansion, $s_\lambda = \sum_{T \in \text{SYT}(\lambda)} F_{n, \text{Des}(T)}$, and the principal specialization of a fundamental quasisymmetric function is, by the theory of P -partitions,

$$F_{n, S}(1, q, q^2, \dots) = \frac{q^{\sum_{i \in S} i}}{(q; q)_n} = \frac{q^{\text{maj}(S)}}{(q; q)_n}.$$

Summing over T and clearing $(q; q)_n$ gives the first equality; the second is the quoted hook-length form. (Both sides are equal as rational functions of q , and the left side is a polynomial.) $\square$

Lemma 3 (Frobenius on the geometric alphabet). *For every $\lambda \vdash n$, as rational functions of q ,*

$$s_\lambda(1, q, q^2, \dots) = \sum_{\mu \vdash n} \frac{\chi^\lambda(\mu)}{z_\mu} \prod_{r \geq 1} (1 - q^r)^{-m_r(\mu)}.$$

Proof. The Frobenius character formula is the expansion $s_\lambda = \sum_{\mu \vdash n} z_\mu^{-1} \chi^\lambda(\mu) p_\mu$ in Λ . Applying the (continuous) specialization that sends the alphabet to X_q and using (2), $p_\mu(X_q) = \prod_r p_r(X_q)^{m_r(\mu)} = \prod_r (1 - q^r)^{-m_r(\mu)}$, gives the claim. $\square$

4 The sieve at $q = -1$

Combining Lemmas 2 and 3,

$$f^\lambda(q) = \sum_{\mu \vdash n} \frac{\chi^\lambda(\mu)}{z_\mu} \underbrace{(q; q)_n \prod_{r \geq 1} (1 - q^r)^{-m_r(\mu)}}_{=: g_\mu(q)}. \quad (3)$$

Each g_μ is a rational function of q ; we determine its behaviour at $q = -1$.

Order of vanishing at $q = -1$. The cyclotomic factorization $q^i - 1 = \prod_{d|i} \Phi_d(q)$ is squarefree, and $\Phi_2(q) = q + 1$ divides $q^i - 1$ if and only if $2 \mid i$, in which case it appears to multiplicity one. Hence

$$\text{ord}_{q=-1}(1 - q^i) = \begin{cases} 1, & i \text{ even}, \\ 0, & i \text{ odd}, \end{cases}$$

where $\text{ord}_{q=-1}$ denotes order of vanishing in $(q + 1)$ (negative for a pole). Consequently

$$\text{ord}_{q=-1}(q; q)_n = \#\{i \in [1, n] : 2 \mid i\} = \left\lfloor \frac{n}{2} \right\rfloor,$$

and, writing $e(\mu) := \sum_{r \text{ even}} m_r(\mu)$ for the number of even parts of μ (with multiplicity),

$$\text{ord}_{q=-1} \prod_{r \geq 1} (1 - q^r)^{-m_r(\mu)} = - \sum_{r \text{ even}} m_r(\mu) = -e(\mu).$$

Therefore

$$\text{ord}_{q=-1} g_\mu = \left\lfloor \frac{n}{2} \right\rfloor - e(\mu). \quad (4)$$

Lemma 4 (Only μ_0 survives). *For every $\mu \vdash n$ one has $e(\mu) \leq \lfloor n/2 \rfloor$, with equality if and only if $\mu = \mu_0 = 2^{\lfloor n/2 \rfloor} 1^{n \bmod 2}$. Consequently g_μ is regular at $q = -1$ for all μ , with $g_\mu(-1) = 0$ unless $\mu = \mu_0$.*

Proof. The even parts of μ are each ≥ 2 , so their total is $\sum_{r \text{ even}} r m_r(\mu) \geq 2e(\mu)$; since this total is at most n , we get $e(\mu) \leq n/2$, i.e. $e(\mu) \leq \lfloor n/2 \rfloor$. Equality forces two things at once: the total of the even parts equals exactly $2\lfloor n/2 \rfloor$, so every even part equals 2; and the odd parts then sum to $n - 2\lfloor n/2 \rfloor = n \bmod 2$, i.e. there are no odd parts if n is even, and a single part 1 if n is odd. Either way $\mu = \mu_0$.

By (4), $\text{ord}_{q=-1} g_\mu = \lfloor n/2 \rfloor - e(\mu) \geq 0$, so each g_μ is regular (no pole) at $q = -1$; and $g_\mu(-1) = 0$ precisely when the order is strictly positive, i.e. when $e(\mu) < \lfloor n/2 \rfloor$, i.e. for all $\mu \neq \mu_0$. $\square$

Lemma 5 (The surviving constant). *Write $a = \lfloor n/2 \rfloor$ and $b = n \bmod 2$. Then*

$$g_{\mu_0}(-1) = \lim_{q \rightarrow -1} (q; q)_n (1 - q^2)^{-a} (1 - q)^{-b} = 2^a a! = z_{\mu_0}.$$

Proof. For $\mu_0 = 2^a 1^b$ we have $\prod_r (1 - q^r)^{-m_r(\mu_0)} = (1 - q^2)^{-a} (1 - q)^{-b}$. Split $(q; q)_n$ into odd- and even-indexed factors:

$$(q; q)_n = \left(\prod_{\substack{1 \leq i \leq n \\ i \text{ odd}}} (1 - q^i) \right) \left(\prod_{k=1}^a (1 - q^{2k}) \right).$$

The number of odd i in $[1, n]$ is $\lfloor n/2 \rfloor = a + b$, and each factor $1 - q^i \rightarrow 1 - (-1) = 2$ as $q \rightarrow -1$, so the first product tends to 2^{a+b} . For the even part,

$$\frac{\prod_{k=1}^a (1 - q^{2k})}{(1 - q^2)^a} = \prod_{k=1}^a \frac{1 - q^{2k}}{1 - q^2} = \prod_{k=1}^a (1 + q^2 + \dots + q^{2(k-1)}) \xrightarrow{q \rightarrow -1} \prod_{k=1}^a k = a!,$$

since $q^2 \rightarrow 1$. Finally $(1 - q)^{-b} \rightarrow 2^{-b}$. Multiplying,

$$g_{\mu_0}(-1) = 2^{a+b} \cdot a! \cdot 2^{-b} = 2^a a!.$$

On the other hand $z_{\mu_0} = 2^a a! \cdot 1^b b! = 2^a a!$ because $b \in \{0, 1\}$ and $b! = 1$. Hence $g_{\mu_0}(-1) = z_{\mu_0}$. $\square$

5 Proof of the Theorem

Proof of Theorem 1. Since f^λ is a polynomial, its value at $q = -1$ is obtained by letting $q \rightarrow -1$ in any equal rational expression. Taking that limit in (3) and using that the sum is finite,

$$f^\lambda(-1) = \sum_{\mu \vdash n} \frac{\chi^\lambda(\mu)}{z_\mu} g_\mu(-1).$$

By Lemma 4, $g_\mu(-1) = 0$ for every $\mu \neq \mu_0$, so only the μ_0 -term survives:

$$f^\lambda(-1) = \frac{\chi^\lambda(\mu_0)}{z_{\mu_0}} g_{\mu_0}(-1) = \frac{\chi^\lambda(\mu_0)}{z_{\mu_0}} \cdot z_{\mu_0} = \chi^\lambda(\mu_0),$$

using Lemma 5. Combined with Lemma 1 ($\sum_T (-1)^{\text{comaj}(T)} = f^\lambda(-1)$) and μ_0 is the cycle type of w_0 , this is exactly

$$\sum_{T \in \text{SYT}(\lambda)} (-1)^{\text{comaj}(T)} = \chi^\lambda(w_0). \quad \square$$

Remark 1 (The sieve is the whole idea). The mechanism is worth naming. The fake degree $f^\lambda(q)$ is a single polynomial that secretly packages *all* character values $\chi^\lambda(\mu)$, weighted by z_μ^{-1} and by the rational “size” $g_\mu(q)$ of the conjugacy class on the geometric alphabet. Evaluating at the primitive second root of unity $q = -1$ switches on the zeros of $(q; q)_n$: there are exactly $\lfloor n/2 \rfloor$ of them, one per even index. A cycle type μ can absorb at most one such zero per even part, so it survives the evaluation only if it carries the maximal possible number $\lfloor n/2 \rfloor$ of even parts — which, because even parts are ≥ 2 , pins μ to the unique type $2^{\lfloor n/2 \rfloor} 1^{n \bmod 2}$. That type is w_0 ’s. The same argument with a primitive d -th root of unity selects, by the identical counting, the cycle types with the maximal number of parts divisible by d ; this is the structural seed of the stretch goal (a graded statement at higher roots of unity).

Remark 2 (Relation to the companion note). The note *The $q \rightarrow -1$ fiber of the branch-exponent bridge* proves, rigorously and independently, that $\chi^\lambda(w_0)$ equals a signed domino count $(-1)^{n(\lambda)} \# \text{SYT}_{\text{dom}}(\lambda)$ (vanishing unless the 2-core of λ is empty for n even, or a single cell for n odd), via the Murnaghan–Nakayama rule on the reversal class. Theorem 1 closes the remaining link in that note’s four-way chain

$$\underbrace{\sum_T (-1)^{s(T)}}_{\text{spectral}} = \underbrace{\sum_T (-1)^{\text{comaj}(T)}}_{\text{descent}} = \underbrace{\chi^\lambda(w_0)}_{\text{character}} = \underbrace{(-1)^{n(\lambda)} \# \text{SYT}_{\text{dom}}(\lambda)}_{\text{domino}},$$

the first equality being the note’s parity reduction $s(T) \equiv \text{comaj}(T) \pmod{2}$. The $q = -1$ fiber of Theorem B is now a complete, self-contained theorem.

6 Computational verification

All claims were checked by independent computation.

- **The full identity.** For every one of the 138 partitions of $n \leq 10$: the left side $\sum_T (-1)^{\text{comaj}(T)}$ (by direct SYT enumeration and descent computation) equals $f^\lambda(-1) = \sum_T (-1)^{\text{maj}(T)}$, and both equal $\chi^\lambda(\mu_0)$ computed by an independent Murnaghan–Nakayama rim-hook recursion. Zero mismatches. (Sanity checks: $\chi^{(n)}(w_0) = 1$ and $\chi^{(1^n)}(w_0) = (-1)^{\lfloor n/2 \rfloor} = \text{sgn}(w_0)$, both confirmed.)
- **Lemma 1.** The multisets $\{\text{maj}(T)\}$ and $\{\text{comaj}(T)\}$ over $\text{SYT}(\lambda)$ coincide for all partitions of $n \leq 10$.
- **Lemma 2.** The polynomial identity $\sum_T q^{\text{maj}(T)} = q^{n(\lambda)}(q; q)_n / \prod_x (1 - q^{h(x)})$ verified symbolically for all partitions of $n \leq 7$.
- **Lemmas 4–5.** For all partitions of $n \leq 8$: $\lim_{q \rightarrow -1} g_\mu(q) = 0$ for $\mu \neq \mu_0$ and $\lim_{q \rightarrow -1} g_{\mu_0}(q) = z_{\mu_0}$ (symbolic limits); and for all $n \leq 10$, μ_0 is the unique partition of n with $e(\mu) = \lfloor n/2 \rfloor$.

Scripts. `~/projects/scratch/2026-06-03-w0-parity-domino/claim1.py` (full identity, $n \leq 10$); the verification driver for this note is recorded alongside it.